

RESEARCH REPORT

Enterprise Customer Domain in Downstream Oil & Gas

Business Capability, MDM, and Bounded-Context Research

EXECUTIVE CONCLUSION

The true enterprise Customer domain is broader than Order-to-Cash. O2C should remain the first proof-of-value bounded context, while Customer/Party identity, roles, relationships, channel networks, loyalty, service and insight are governed across multiple contexts.

Initiative	Enterprise Performance Model
Workstream	Domain Modeling, with KPI Store and Data Governance implications
Status	Research-backed recommendation - not an approved baseline change
Prepared	21 August 2026

Executive summary

Research conclusion. Yes - the enterprise Customer domain is broader than O2C. The prior research found that three architectural notions had been collapsed into one name:

1. Business-capability / subject domain - acquisition, relationship management, transactions, dealer and retail networks, loyalty, service, channels and customer intelligence.
2. Data/MDM domain - a Party-centered foundation of Person, Organization, roles, relationships, identities, accounts, hierarchies, contact points, classifications, consent/preferences, locations and cross-system equivalences.
3. O2C bounded context / value stream - the context in which Party is specialized into Sold-To, Bill-To, Payer, Ship-To/Delivery Point, consignee, credit counterparty and loading-authorized meanings.

DECISION FRAMING

Retain O2C as proof of value 1. Change the architectural wording from “Customer domain = O2C” to “Enterprise Customer/Party domain supports multiple bounded contexts; Commercial/O2C is the first modeled context.”

Status and evidence handling

- Known current EPM lock: “Customer is modeled through order-to-cash.” This is an approved project lock and is not changed by this report.
- Validated project direction: the sponsor handoff describes a “Customer / order-to-cash reusable pattern (proof of value 1).”
- Research evidence: primary-source process, MDM, downstream operator, SAP, PIDX and DDD materials consistently support a broader enterprise scope.
- Recommendation: refine, do not silently override, the lock through an Architecture Decision Log entry and affected artifact updates.

1. Synthesized definitions

Enterprise Customer business domain

The Customer domain comprises the capabilities through which an enterprise identifies, establishes, governs, develops, transacts with, serves and understands parties that currently or potentially consume, purchase, receive, pay for, distribute, resell or otherwise participate in its market offerings.

In Downstream Oil & Gas, this includes wholesalers, distributors, dealers, branded-site operators, rack customers, truck-delivered end users, airlines, aviation dealers, marine customers, vessel owners/operators, utilities, traders, petroleum companies, lubricant distributors, retail consumers, loyalty members and fleet-card participants. The same legal entity may act in several roles. [S08-S11]

Enterprise Customer data/MDM domain

The Customer data domain is the governed set of persistent Party-centered master and reference information needed to recognize and relate those actors consistently across business contexts.

DURABLE CORE

Party -> Person / Organization -> Identity -> Role -> Relationship -> Account -> Hierarchy -> Contact / Location -> Classification -> Preference / Consent -> Contextual assignments

This is materially broader than Customer -> Order -> Invoice. IBM explicitly includes relationships, equivalencies, identifiers, financial profiles, privacy, campaigns, interactions, compliance, grouping and hierarchy. Salesforce, Reltio and Informatica similarly treat Customer through Party/entity structures and contextual roles. [S03-S05, S12-S15]

O2C bounded context

O2C is the commercial transaction context in which a subset of Customer/Party meaning is specialized for quote, agreement, pricing, credit, order, fulfilment, delivery, invoicing, receivables, collections and disputes.

RELATIONSHIP

Enterprise Customer/Party domain -> multiple bounded contexts -> Commercial / Order-to-Cash is one of them. Customer domain does not equal O2C.

2. Process-framework interpretation

APQC does not define one monolithic Customer domain. The Downstream Petroleum PCF spans manufacturing, marketing, trading and distribution, while customer-related work appears across market/sell, fulfilment, finance and customer-service categories. APQC separately identifies Manage Customer Service after delivery. [S01-S02]

A process architecture therefore distinguishes Market/Acquire -> Manage Account -> Sell -> Fulfil -> Bill/Collect -> Serve/Retain. Customer is a subject/capability domain; O2C is an end-to-end value stream. They are different classification dimensions.

3. Comparison across universal and MDM treatments

Source	Fundamental treatment	Meaning of Customer	Beyond O2C	EPM implication	Evidence
Silverston / Universal Data Models	Party, Person, Organization, roles and relationships are foundational; orders, shipments and invoices are separate subject areas.	Customer is a contextual party role.	Separates durable party structures from transaction structures.	Model O2C roles as projections of a stable Party core.	S16
IBM InfoSphere MDM	Party domain.	Party covers customers, vendors and suppliers; a party can play different roles in different contexts.	Relationships, equivalencies, identifiers, financial profile, privacy, campaigns, interactions, compliance, grouping and hierarchy.	Enterprise Customer MDM is cross-process.	S03-S05
SAP S/4HANA / RFNO	Business Partner and time-valid relationship roles.	Customer and payment/fleet/network roles are contextual uses of a partner.	Dealer, location manager, owner, card payer/co-user, issuer, clearing house and network operator coexist.	Keep O2C partner functions separate from durable party/network relationships.	S06
Reltio	Entities with configurable roles.	Customer, Prospect and Vendor can be roles assigned to companies and individuals.	One entity can carry multiple business roles.	Customer should not be the universal root entity.	S14

Source	Fundamental treatment	Meaning of Customer	Beyond O2C	EPM implication	Evidence
Salesforce Data 360	Party connected to Individual, Organization, Account and contextual DMOs.	Party represents contacts/clients; related models express consent, loyalty and payment contexts.	Leads, consents, loyalty members, contact points and payment methods connect to Party.	CRM, privacy and loyalty sit outside O2C while sharing identity.	S12-S13
Informatica Customer 360	Customer domain implemented through business entities.	A Customer domain may be represented by Person, Patient or Policy Holder entities.	Supports multiple entity types, relationships and mastered records.	Use a multidimensional customer master rather than one O2C record.	S15
DAMA-DMBOK	Master/reference data governance as an enterprise discipline.	Does not prescribe one Customer entity pattern in public overview material.	Provides governance vocabulary and cross-domain management principles.	Use as governance authority, not sole proof of a Customer model.	S17
DDD / Fowler	Different bounded contexts may model the same term differently.	Customer can be polysemous across business functions.	Context maps relate models without forcing one giant aggregate.	Do not collapse O2C, loyalty, service, dealer and trading meanings.	S18-S19

4. Downstream-specific evidence

Wholesale/rack and branded channels

Valero distinguishes truck-rack purchasers, branded distributors/dealers and bulk customers such as petroleum companies, traders, airlines and utilities. Phillips 66 distinguishes marketers, branded/unbranded outlets, consignment, aviation dealers, retail joint ventures, marine bunker fuels and lubricants. A single “Wholesale” category cannot represent these relationship structures. [S08-S10]

Retail network structure

SAP RFNO distinguishes company-owned/company-operated, company-owned/dealer-operated and dealer-owned/dealer-operated patterns, with separate location, operator, owner, card and settlement roles. These are durable network relationships, not just O2C partner functions. [S06]

Fleet cards and settlement ecosystems

RFNO includes Dealer, Location Manager, Service Station Owner, Payment Card Owner, Payment Card Payer, Co-user, Driver, Clearing House, Issuer and Network Operator. Relationships can be time-valid and settlement-relevant. [S06]

Aviation, marine and lubricants

Operator disclosures separate aviation dealers, marine customers, commercial fuel buyers and lubricant channels. These contexts involve airlines, dealers, into-plane agents, vessel actors, bunker buyers, marketers and distributors. [S09-S10]

Loyalty and retail consumers

Rewards+ introduces member identity, app/card/Alt-ID, points, tier status, payment linkage and offers. A motorist who is not material to wholesale O2C becomes first-class in loyalty and consumer engagement. [S11-S13]

Terminal and lift ecosystem

PIDX publishes LoadingDeliveryReceipt, planned movement, terminal messages and RighttoLift schemas. This supports treating Loading-Authorized Party as a distinct contextual meaning rather than generic Customer. [S07]

5. Capability boundary - 30 capabilities

Classification: A = O2C core; B = broader Customer lifecycle; C = shared Party/Customer MDM or adjacent capability. Combined labels identify explicit domain interfaces, not ambiguity.

#	Capability	Class	Boundary rationale	Evidence
1	Party / Person / Organization master	C	Persistent identity exists independently of a sale.	S03-S05, S14, S16
2	Identity resolution, duplicate matching and cross-system equivalence	C	Enterprise recognition is not a sales transaction.	S03
3	Party roles and relationship management	C	Roles vary by context and time.	S04-S06, S14
4	Customer account master	C	Account is reused by sales, service, loyalty and payment contexts.	S12
5	Address, contact point and communication identity	C	Shared across sales, service, marketing, logistics and compliance.	S03, S12, S16
6	Corporate/customer hierarchy and grouping	C	Multiple functions consume hierarchies independently of orders.	S03, S16
7	Classification, segment and channel assignment	C	Used by marketing, sales, service, analytics and channel management.	S03, S12
8	Lead and prospect management	B	Precedes O2C.	S12, S14
9	Marketing campaign / audience management	B	Acquisition and engagement occur outside order execution.	S03
10	Privacy, consent and communication preference	B/C	Customer-lifecycle capability with shared-data implications.	S03, S12
11	Loyalty membership, tier and rewards	B	Distinct consumer-engagement lifecycle.	S11, S13
12	Retail consumer digital identity / app / payment linkage	B	Identity persists beyond an individual sale.	S11-S13
13	Dealer / franchise / retail-network partner management	B	Governs channel participants and sites.	S06, S08-S09
14	Retail-site operating model - COCO/CODO/DODO	B	Ownership and operation relationships outlive transactions.	S06
15	Brand, wholesaler, marketer/jobber and distributor relationships	B	Channel governance rather than order execution.	S08-S10
16	Quote / commercial offer	A	Direct commercial transaction precursor.	S16
17	Contract / agreement / volume commitment / rebate	A	Governs terms under which O2C transactions occur.	S16
18	Customer-specific pricing and commercial terms	A	Creates and prices the sale.	S16
19	Credit limit / exposure / guarantee / block	A + Finance	O2C performs credit decisions while consuming governed finance/party data.	S03
20	Sales order capture and change	A	Core O2C commitment and transaction management.	S16
21	Allocation, nomination and fulfilment scheduling	A + Supply	O2C commits demand; supply/logistics contexts plan execution.	S01, S07

#	Capability	Class	Boundary rationale	Evidence
22	Right-to-lift and loading authorization	C + Terminal	A shared authorization context determines who may lift; it is not a generic customer role.	S07
23	Delivery, shipment and proof of delivery	A + Logistics	Fulfilment belongs to the value stream but uses logistics assets and events.	S01, S07, S16
24	Billing, invoicing and tax determination	A + Tax	Core O2C outcome with shared tax/location master inputs.	S16
25	Receivables and cash application	A + Finance	Financial settlement of O2C transactions.	S16
26	Collections and dunning	A + Finance	O2C close-stage activity, governed with Finance.	S02
27	Dispute, deduction and claims resolution	A + Service/Finance	Triggered by O2C but may span service and finance contexts.	S02
28	Customer inquiry and service-case management	B	Post-delivery service is a distinct process area in APQC.	S02
29	Complaint, feedback, warranty and recall management	B	Customer service lifecycle, not transaction execution.	S02
30	Customer analytics, profitability, lifetime value and next-best action	B/C	Cross-context insight uses mastered identity and events from many processes.	S03, S12

6. EPM implications and recommendations

Preserve the pilot boundary

Continue O2C as the first proof of value because it is bounded, operationally testable and already aligned with the sponsor handoff.

Refine the lock through governance

Propose an ADL decision that distinguishes the enterprise Customer/Party domain from the O2C bounded context. Do not edit the approved lock silently.

Adopt a Party-centered conceptual core

Use Party, Person, Organization, Identity, Role, Relationship, Account, Hierarchy, Contact/Location and contextual assignments as the reusable enterprise foundation.

Keep context-specific meanings explicit

Model Sold-To, Bill-To, Payer, credit counterparty, consignee and Loading-Authorized Party in their proper contexts. Preserve the current EPM conclusion that Ship-To is a location/delivery point where applicable, not automatically a party role.

Separate network governance from transactions

Create a Retail/Dealer Network bounded context for ownership, operation, brand, supply, fleet-card and settlement relationships.

Connect KPIs by context

Map customer-related KPIs to the accountable objective, process and bounded context. Do not infer an enterprise Customer KPI merely because a calculation uses a customer field.

Govern cross-context identifiers and lineage

Define source-system equivalences, survivorship, role validity, hierarchy effective dates, consent, tax/license attributes and lineage to Power BI and future KPI Store consumption.

Pilot explicit interfaces

For the O2C proof of value, document which Party/Customer master attributes are mastered outside O2C, which are projected into O2C and which events flow back.

7. Assumptions, limitations and open issues

Assumptions

- The report is industry-generic and does not assert one company's operating model.
- The source set provides architectural evidence, not approval of a specific EPM implementation.
- O2C remains the near-term modeling and delivery priority.
- Customer is treated as a contextual role/subject over a durable Party foundation; local systems may use different labels.

Limitations

- The immediately preceding ChatGPT response was recoverable only through capability 19 because the thread interface truncates long individual messages. The missing tail was reconstructed from the same cited primary-source logic and is presented as recommendation.
- Public SAP, PIDX and operator materials show patterns and channel complexity but do not define the EPM target-state model.
- DAMA public material supports governance positioning but does not prescribe a specific Customer entity pattern.

Open issues

- Should the approved lock be reworded now, or after the O2C proof of value validates the reusable pattern?
- Where will enterprise Party identity, role validity and hierarchy stewardship sit organizationally?
- Which contexts are next after O2C: Retail/Dealer Network, Loyalty/Consumer, Customer Service or Trading/Exchange Partner?
- Which tax, excise, license, consent and payment attributes belong in shared master data versus context-owned data products?
- Which candidate Customer KPIs require enterprise identity resolution versus O2C-only dimensions?

8. Artifact Update Block

Artifact update field	Report outcome
Conclusions	Enterprise Customer is broader than O2C. O2C is the first bounded context/proof of value, not the complete domain.
Decisions and status	Recommendation only. No approved baseline has been changed. An ADL decision is required before changing the current lock.
Definitions added or changed	Proposed definitions for Enterprise Customer business domain, Customer data/MDM domain and O2C bounded context; A/B/C capability classification.
Assumptions	Industry-generic scope; Party-centered conceptual core; O2C remains first priority.
Open questions and conflicts	The approved lock “Customer is modeled through order-to-cash” conflicts with broader research evidence unless interpreted as pilot scope.
Source evidence	APQC, IBM, SAP, PIDX, operator filings/material, Salesforce, Reltio, Informatica, Silverston/Wiley, DAMA and DDD/Fowler; exact URLs in Appendix A.
Artifacts to create or update	Architecture Decision Log; EPM Master Index; Business Glossary and Measurement Taxonomy; Domain Model Repository; Downstream Business Architecture; Data Product Portfolio; Source and Traceability Register; Open Issues and Assumptions Log.
Suggested version/status changes	Keep current approved artifacts unchanged. Create a Candidate ADL entry and Candidate domain-scope note; update canonical artifacts only after approval.
Next action	Hold a focused architecture/governance review to approve the domain-versus-context distinction and define the O2C interface to Party/Customer master data.

Appendix A - Exact source URLs

The URLs below are the exact primary or first-party pages used to ground the reconstructed report. Access dates and product versions should be refreshed before formal approval.

S01 APQC - Downstream Petroleum PCF v7.2.2

<https://www.apqc.org/resource-library/resource-listing/apqc-process-classification-framework-pcf-downstream-petroleum-2>

S02 APQC - Understanding the PCF Elements

<https://www.apqc.org/sites/default/files/files/PCF%20Collateral/Understanding%20the%20PCF%20Elements%20%20-%20FINAL.pdf>

S03 IBM - Introduction to the Party domain

<https://www.ibm.com/docs/en/imdm/11.6.0?topic=dpm-party-domain>

S04 IBM - Party roles

<https://www.ibm.com/docs/en/imdm/14.0.0?topic=domain-party-roles>

S05 IBM - Party domain model

<https://www.ibm.com/docs/en/imdm/14.0.0?topic=overview-party-domain-model>

S06 SAP - S/4HANA Supply Chain for Secondary Distribution - RFNO guide

https://help.sap.com/doc/4b18d62bc99d44ca87e662c045a872ff/4.0.2/en-US/S4SCSD_40_EN.pdf

S07 PIDX - Standards and complete downstream library

<https://pidx.org/standards/>

S08 Valero - 2025 Form 10-K

<https://www.sec.gov/Archives/edgar/data/1035002/000162828026011499/vlo-20251231.htm>

S09 Phillips 66 - 2025 Form 10-K

<https://www.sec.gov/Archives/edgar/data/1534701/000153470126000006/psx-20251231.htm>

S10 ExxonMobil - 2017 Financial and Operating Review

<https://corporate.exxonmobil.com/-/media/Global/Files/annual-report/2017-Financial-and-Operating-Review.pdf>

S11 Exxon Mobil - Rewards+ program Q&A

https://www.exxon.com/-/media/project/wep/exxon/exxon-retail-us/updated-pdfs/the-exxon-mobil-rewards-program-qa.pdf?hash=f55388f17adf51a01e0da83261c4359d&sc_lang=en

S12 Salesforce - Party DMO

<https://developer.salesforce.com/docs/data/data-cloud-dmo-mapping/guide/c360dm-party-dmo.html>

S13 Salesforce - Loyalty Program Member DMO

https://developer.salesforce.com/docs/atlas.en-us.c360a_api.meta/c360a_api/c360dm_loyalty_program_member_dmo.htm

S14 Reltio - Entity roles

<https://docs.reltio.com/en/reltio/what-does-reltio-do/what-reltio-does-at-a-glance/data-unification-and-mdm-at-a-glance/data-unification-and-mdm-in-detail/reltio-information-model/lookups-for-attribute-code-values/entity-roles>

S15 Informatica - Customer 360 business entities

<https://docs.informatica.com/master-data-management/customer-360/10-3/user-guide/introduction-to-informatica-mdm---customer-360/key-concepts/business-entities.html>

S16 Wiley - Universal Data Models excerpt

<https://catalogimages.wiley.com/images/db/pdf/9780471380238.excerpt.pdf>

S17 DAMA International - DAMA-DMBOK overview

<https://www.dama.org/cpages/body-of-knowledge>

S18 Martin Fowler - Bounded Context

<https://martinfowler.com/bliki/BoundedContext.html>

S19 Martin Fowler - Polyseme

<https://martinfowler.com/bliki/Polyseme.html>

Appendix B - Project-source traceability

Item	Project source	Relevant evidence	Treatment
Current lock	sources/EPM-LOCK-2026-08-16.md	“Customer is modeled through order-to-cash.”	Approved baseline; not changed.
Domain approach	sources/EPM-MOD-000-approach-domain-analysis-v0.3.html	Customer domain “feeds Order-to-Cash”; contextual role and fulfilment distinctions.	Higher-precedence internal baseline.
Sponsor handoff	sources/sponsor-handoff-deliverables.md	“Customer / order-to-cash reusable pattern (proof of value 1).”	Supports interpreting O2C as pilot context.
Controlled vocabulary	sources/production-ready-ontology-01-controlled-vocabulary(1).md	Do not collapse Sold-To, Payer, Bill-To, Ship-To and Loading-Authorized Party into “Customer.”	Established working evidence.